

# PROS6-A1250-OSB

Serie PROS · R513A · senza serbatoio / without tank



PROS6-A1250-OSB

RESA FRIGORIFERA · COOLING CAPACITY

**820** kW

EER · 12/7 °C · AIR 35 °C

**2,96**

RESA NOMINALE · 15 °C · AIR 25 °C

**1.247** kW

PORTATA ACQUA · WATER FLOW

**141,1** m<sup>3</sup>/h

|                            |                 |                                        |                        |
|----------------------------|-----------------|----------------------------------------|------------------------|
| Compressori / Compressors  | 2 × vite        | L × P × H (mm)                         | 11.415 × 2.197 × 2.402 |
| Circuiti / Circuits        | 2               | Serbatoio / Tank                       | —                      |
| Ventilatori / Fans         | 20              | Taglia / Size                          | 1250                   |
| Refrigerante / Refrigerant | R513A · GWP 631 | Campo aria esterna / Ambient air range | -5 / +50 °C            |

## Resa in sintesi / Capacity at a glance <sup>kW</sup>

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C | 50 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 914   | 868   | 820   | 771   | 721   | 671   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 1.247 | 1.186 | 1.124 | 1.060 | 995   | 929   |

Refrigeratori centralizzati condensati ad aria con compressori a vite, per le grandi potenze industriali e commerciali, con parzializzazione della potenza 25-50-75-100 %.

Centralised air-cooled chillers with screw compressors for large industrial and commercial loads, with 25-50-75-100 % capacity steps.

# Dati tecnici

## Technical data

PROS6 - A1250 - OSB

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.246,76 kW              |
| Resa frigorifera / Cooling capacity | 1.072.214 kcal/h         |
| Potenza assorbita / Power input     | 235,66 kW                |
| EER / EER                           | 5,29                     |
| Portata acqua / Water flow          | 214,44 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 820,18 kW                |
| Potenza assorbita / Power input     | 276,87 kW                |
| EER / EER                           | 2,96                     |
| ESEER / ESEER                       | 3,94                     |
| Portata acqua / Water flow          | 141,07 m <sup>3</sup> /h |

### Componenti / Components

|                                             |                      |
|---------------------------------------------|----------------------|
| Compressori / Compressors                   | 2 × Vite             |
| Circuiti frigoriferi / Refrigerant circuits | 2                    |
| Ventilatori / Fans                          | 20                   |
| Condensatore / Condenser                    | a microcanali        |
| Evaporatore / Evaporator                    | a fascio tubiero     |
| Refrigerante / Refrigerant                  | R513A • GWP 631 • A1 |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | 11.415 × 2.197 × 2.402 mm |
| Campo aria esterna / Ambient air range                                                    | -5 / +50 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 • 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 803,8 kW     |
| MEG 30 % | × 0,950 | 779,2 kW     |
| MEG 40 % | × 0,920 | 754,6 kW     |
| MPG 20 % | × 0,975 | 799,7 kW     |
| MPG 30 % | × 0,960 | 787,4 kW     |
| MPG 40 % | × 0,950 | 779,2 kW     |

# Prestazioni Performance

PROS6-A1250-OSB · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15    | 20    | 25    | 30    | 35    | 40    | 45    | 50    |
|---------------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1             | 785   | 750   | 713   | 676   | 637   | 597   | 557   | 516   |
| 2             | 819   | 782   | 745   | 706   | 666   | 625   | 583   | 540   |
| 3             | 853   | 815   | 776   | 736   | 694   | 652   | 608   | 564   |
| 4             | 889   | 850   | 809   | 768   | 725   | 681   | 636   | 590   |
| 5             | 925   | 885   | 843   | 800   | 755   | 709   | 663   | 616   |
| 6             | 964   | 922   | 879   | 834   | 788   | 740   | 692   | 643   |
| 7             | 1.002 | 959   | 914   | 868   | 820   | 771   | 721   | 671   |
| 8             | 1.043 | 999   | 952   | 904   | 855   | 804   | 752   | 700   |
| 9             | 1.084 | 1.038 | 990   | 940   | 889   | 837   | 784   | 729   |
| 10            | 1.128 | 1.080 | 1.030 | 979   | 926   | 872   | 817   | 760   |
| 11            | 1.172 | 1.122 | 1.071 | 1.018 | 963   | 907   | 850   | 792   |
| 12            | 1.218 | 1.167 | 1.113 | 1.059 | 1.002 | 944   | 885   | 825   |
| 13            | 1.264 | 1.211 | 1.156 | 1.100 | 1.041 | 981   | 920   | 858   |
| 14            | 1.313 | 1.258 | 1.201 | 1.143 | 1.083 | 1.021 | 958   | 893   |
| 15            | 1.362 | 1.305 | 1.247 | 1.186 | 1.124 | 1.060 | 995   | 929   |
| 16            | 1.414 | 1.355 | 1.295 | 1.232 | 1.168 | 1.102 | 1.035 | 967   |
| 17            | 1.466 | 1.405 | 1.343 | 1.279 | 1.212 | 1.144 | 1.075 | 1.004 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  | 50  |
|---------------|-----|-----|-----|-----|-----|-----|-----|-----|
| 1             | 188 | 206 | 226 | 248 | 272 | 299 | 328 | 360 |
| 2             | 189 | 207 | 227 | 249 | 273 | 300 | 329 | 362 |
| 3             | 190 | 208 | 227 | 249 | 274 | 301 | 330 | 363 |
| 4             | 190 | 208 | 228 | 250 | 275 | 302 | 331 | 364 |
| 5             | 191 | 209 | 229 | 251 | 275 | 302 | 332 | 365 |
| 6             | 192 | 210 | 230 | 252 | 276 | 303 | 333 | 366 |
| 7             | 192 | 210 | 230 | 252 | 277 | 304 | 334 | 367 |
| 8             | 193 | 211 | 231 | 253 | 278 | 305 | 335 | 368 |
| 9             | 194 | 212 | 231 | 254 | 278 | 306 | 336 | 369 |
| 10            | 195 | 212 | 232 | 254 | 279 | 306 | 337 | 370 |
| 11            | 195 | 213 | 233 | 255 | 280 | 307 | 337 | 371 |
| 12            | 196 | 214 | 233 | 256 | 280 | 308 | 338 | 372 |
| 13            | 197 | 214 | 234 | 256 | 281 | 309 | 339 | 373 |
| 14            | 198 | 215 | 235 | 257 | 282 | 309 | 340 | 374 |
| 15            | 198 | 216 | 236 | 258 | 282 | 310 | 341 | 374 |
| 16            | 199 | 217 | 236 | 258 | 283 | 311 | 341 | 375 |
| 17            | 200 | 218 | 237 | 259 | 284 | 311 | 342 | 376 |

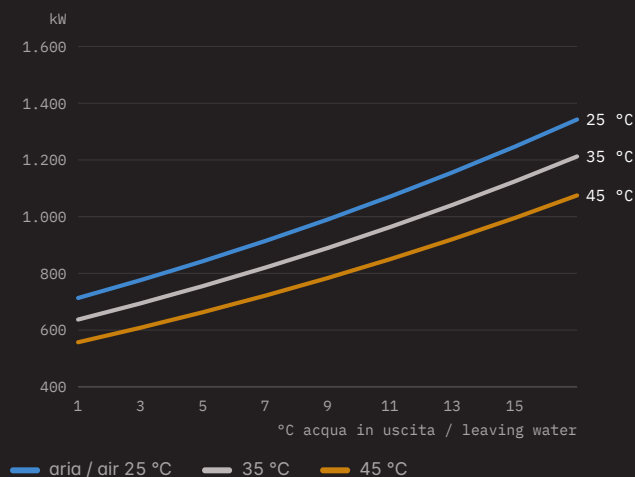
## EER / EER (kW/kW)

| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   | 50   |
|---------------|------|------|------|------|------|------|------|------|
| 1             | 4,17 | 3,63 | 3,15 | 2,72 | 2,34 | 2,00 | 1,70 | 1,43 |
| 2             | 4,33 | 3,78 | 3,28 | 2,84 | 2,44 | 2,08 | 1,77 | 1,49 |
| 3             | 4,49 | 3,92 | 3,41 | 2,95 | 2,54 | 2,17 | 1,84 | 1,56 |
| 4             | 4,67 | 4,08 | 3,55 | 3,07 | 2,64 | 2,26 | 1,92 | 1,62 |
| 5             | 4,84 | 4,23 | 3,68 | 3,19 | 2,74 | 2,35 | 2,00 | 1,69 |
| 6             | 5,03 | 4,40 | 3,83 | 3,31 | 2,85 | 2,44 | 2,08 | 1,76 |
| 7             | 5,21 | 4,56 | 3,97 | 3,44 | 2,96 | 2,54 | 2,16 | 1,83 |
| 8             | 5,40 | 4,73 | 4,13 | 3,57 | 3,08 | 2,64 | 2,25 | 1,90 |
| 9             | 5,60 | 4,91 | 4,28 | 3,71 | 3,20 | 2,74 | 2,33 | 1,97 |
| 10            | 5,80 | 5,09 | 4,44 | 3,85 | 3,32 | 2,85 | 2,43 | 2,05 |
| 11            | 6,00 | 5,27 | 4,60 | 3,99 | 3,44 | 2,95 | 2,52 | 2,13 |
| 12            | 6,21 | 5,46 | 4,77 | 4,14 | 3,57 | 3,07 | 2,62 | 2,22 |
| 13            | 6,43 | 5,65 | 4,94 | 4,29 | 3,70 | 3,18 | 2,71 | 2,30 |
| 14            | 6,65 | 5,85 | 5,11 | 4,45 | 3,84 | 3,30 | 2,82 | 2,39 |
| 15            | 6,86 | 6,05 | 5,29 | 4,61 | 3,98 | 3,42 | 2,92 | 2,48 |
| 16            | 7,09 | 6,25 | 5,48 | 4,77 | 4,13 | 3,55 | 3,03 | 2,58 |
| 17            | 7,32 | 6,46 | 5,66 | 4,93 | 4,27 | 3,68 | 3,14 | 2,67 |

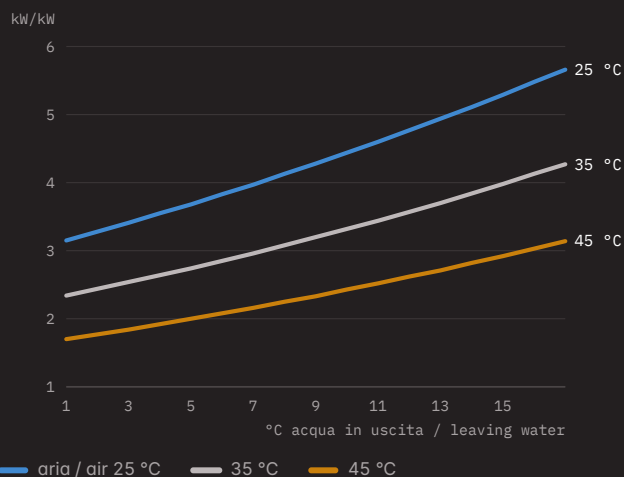
# Curve e dimensioni

## Curves and dimensions

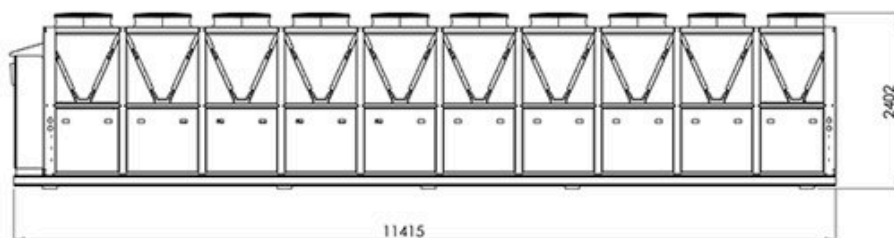
### Resa frigorifera / Cooling capacity



### EER / EER



### Dimensioni / Dimensions · L × P × H 11.415 × 2.197 × 2.402 mm



Disegno indicativo, quote in mm (±10 mm). Dimensioni da confermare in fase d'ordine. / Indicative drawing, dimensions in mm (±10 mm), to be confirmed at order stage.

# Optional — 1/2 Options

PROS6-A1250-OSB · 60 optional disponibili per questa configurazione · 60 options available for this configuration

## Pompe / Pumps

- Pompa 161 m³/h – prevalenza 25 m

Pump 161 m³/h – head 25 m
- Pompa 161 m³/h – prevalenza 40 m

Pump 161 m³/h – head 40 m
- Pompa doppia montata 161 m³/h – prevalenza 25 m

Twin pump, factory -fitted, 161 m³/h – head 25 m
- Pompa doppia montata 161 m³/h – prevalenza 40 m

Twin pump, factory -fitted, 161 m³/h – head 40 m

## Serbatoi e vasi di espansione / Tanks and expansion vessels

- Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata

External water tank 3000 l, AISI 304 with factory -fitted twin AISI 304 pump
- Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata

External water tank 5000 l, AISI 304 with factory -fitted twin AISI 304 pump
- Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata

External water tank 3000 l, AISI 304 with factory -fitted twin pump
- Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata

External water tank 5000 l, AISI 304 with factory -fitted twin pump
- Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 2000 l, AISI 304 with factory -fitted single AISI 304 pump
- Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 3000 l, AISI 304 with factory -fitted single AISI 304 pump
- Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 5000 l, AISI 304 with factory -fitted single AISI 304 pump
- Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola montata

External water tank 2000 l, AISI 304 with factory -fitted single pump
- Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata

External water tank 3000 l, AISI 304 with factory -fitted single pump
- Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata

External water tank 5000 l, AISI 304 with factory -fitted single pump
- Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa doppia montata

External water tank 3000 l, pressurised, galvanised steel with factory -fitted twin pump
- Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa doppia montata

External water tank 5000 l, pressurised, galvanised steel with factory -fitted twin pump
- Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 3000 l, pressurised, galvanised steel with factory -fitted single pump

- Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 5000 l, pressurised, galvanised steel with factory -fitted single pump
- Vaso di espansione 100 l

Expansion vessel 100 l
- Vaso di espansione 200 l

Expansion vessel 200 l
- Vaso di espansione 300 l

Expansion vessel 300 l
- Vaso di espansione 500 l

Expansion vessel 500 l
- Vaso di espansione 750 l

Expansion vessel 750 l

## Valvole e filtri / Valves and filters

- Elettrovalvola 2"

Solenoid valve 2"
- Elettrovalvola 3"

Solenoid valve 3"
- Elettrovalvola 4"

Solenoid valve 4"
- Elettrovalvola 5"

Solenoid valve 5"
- Elettrovalvola 6"

Solenoid valve 6"
- Filtro a dischi in acciaio 2"

Steel disc filter 2"
- Filtro a dischi in acciaio 3"

Steel disc filter 3"
- Filtro a dischi in acciaio 4"

Steel disc filter 4"
- Filtro meccanico a Y 2"

Y -strainer 2"
- Filtro meccanico a Y 3"

Y -strainer 3"
- Filtro meccanico a Y 4"

Y -strainer 4"
- Filtro meccanico a Y 5"

Y -strainer 5"
- Filtro meccanico a Y 6"

Y -strainer 6"
- Valvola a 3 vie 2" miscelatrice

3 -way mixing valve 2"
- Valvola a 3 vie 3" miscelatrice

3 -way mixing valve 3"

# Optional — 2/2

## Options

PROS6-A1250-OSB · 60 optional disponibili per questa configurazione · 60 options available for this configuration

### Valvole e filtri / Valves and filters

Valvola a 3 vie 4" miscelatrice  
3-way mixing valve 4"

Valvola a 3 vie 5" miscelatrice  
3-way mixing valve 5"

Valvola a 3 vie 6" miscelatrice  
3-way mixing valve 6"

Valvola a 3 vie 2" on/off  
3-way valve 2" on/off

Valvola a 3 vie 3" on/off  
3-way valve 3" on/off

Valvola a 3 vie 4" on/off  
3-way valve 4" on/off

Valvola a 3 vie 5" on/off  
3-way valve 5" on/off

Valvola a 3 vie 6" on/off  
3-way valve 6" on/off

### Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori  
Compressor acoustic enclosure

Compressore inverter  
Inverter compressor

Desurriscaldatore a piastre  
Plate desuperheater

Rilevatore fughe gas  
Refrigerant leak detector

### Free cooling / Free cooling

Free cooling 800 kW – 16 ventilatori (rif. 14)  
Free cooling 800 kW – 16 fans (ref. 14)

### Ventilazione e clima / Fans and climate

Inverter ventilatori  
Fan inverter

Ventilatori EC  
EC fans

### Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)  
Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)  
RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)  
RS485 to BACnet communication (cable not included)

### Quadro elettrico / Electrical panel

Analizzatore di rete  
Power analyser

Kit invernale (inverter ventilatori + resistenza quadro + resistenza evaporatore + termostato) fino a -25 °C

Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla  
Spring anti-vibration mounts

Antivibranti in gomma  
Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
- Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
- Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
- Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
- Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
- With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
- Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
- Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
- Procryon reserves the right to change data and features without notice, as part of continuous product improvement.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

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Continuity by design